DATA ANALYSIS AND VISUALIZATION

PROJECT REPORT

(Project Semester January-April 2026)

Traffic Crash AnalysisPortal

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DECLARATION

I, **Vinay Jaiswal** student of **Computer science and Engineering** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

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CERTIFICATE

This is to certify that Vinay Jaiswal bearing Registration no. 12411819 has completed INT387 project titled, “Crime Insights & Intelligence Portal” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor
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Date: 18-04-2026

Acknowledgement

I would like to express my sincere gratitude to my faculty, Dr. Karandeep Kaur, for her continuous guidance, valuable feedback, and consistent support throughout this project.

I would also like to thank Lovely Professional University for providing the resources and academic environment that allowed me to complete this project successfully.

Last but not least, I extend my thanks to my friends for their encouragement and support during the course of this work.

Vinay Jaiswal

Table of Contents

Introduction..... 6
Source of Dataset 7
Dataset Preprocessing..... 8
3.1 Handling Missing Values
3.2 Data Cleaning
3.3 Feature Engineering
3.4 Merging Datasets
3.5 Data Standardization
Analysis on Dataset..... 10
4.1 DRIVER FAULT ANALYSIS
4.2 VEHICLE ANALYSIS
4.3 VEHICLE DAMAGE EXTENT ANALYSIS
4.4 COLLISION TYPE ANALYSIS
4.5 WEATHER AND SURFACE CONDITION ANALYSIS
4.6 TEMPORAL TREND ANALYSIS
4.7 SPEED LIMIT ANALYSIS
4.8 GEOGRAPHIC ANALYSIS
Conclusion 23
Future Scope 25
Reference 27

1. Introduction

1.1 Background

Traffic safety is a critical concern globally, with millions of accidents occurring annually, resulting in significant loss of life and economic burden. Understanding the factors contributing to traffic collisions is essential for developing effective safety interventions and policies. This project focuses on analyzing driver fault and vehicle impact patterns in traffic collisions to identify key risk factors and trends.

1.2 Problem Statement

Traffic collisions result from a complex interplay of driver behavior, vehicle characteristics, environmental conditions, and road infrastructure. By systematically analyzing crash data, we can identify:

- The prevalence of driver fault in collisions
- Vehicle makes most vulnerable to accidents
- Environmental and road conditions that increase crash risk
- Temporal and geographic patterns of collisions
- Relationship between speed limits and crash frequency

1.3 Objectives

The primary objectives of this analysis are:

i. To determine the proportion of crashes attributable to driver fault versus other factors
ii. To identify vehicle makes and models most frequently involved in crashes
iii. To analyze crash patterns based on vehicle damage extent
iv. To examine the relationship between weather conditions, surface conditions, and collision types
v. To identify temporal trends in crash frequency and severity
vi. To analyze the correlation between speed limits and crash volume
vii. To provide geographic distribution of crashes

1.4 Significance

This analysis will provide valuable insights for:

- Insurance companies to assess risk profiles
- Vehicle manufacturers to understand safety vulnerabilities
- Traffic authorities to implement targeted safety measures
- Policy makers to develop evidence-based regulations
- Road users to understand high-risk conditions

2. Source of Dataset

The dataset used in this project was sourced from the Catalog, an open platform that provides real-world datasets for data analysis and visualization practice.

Dataset Name: **Crash Reporting - Drivers Data**

Source Link: https://catalog.data.gov/dataset/crash-reporting-drivers-data

The dataset consists of several fields that capture detailed **road traffic crash–related information**, providing comprehensive insights into crash patterns, driver behavior, vehicle impact, and environmental conditions. The key components of the dataset include:

• **Crash Details:** Core crash-level information such as crash date and time, crash year, collision type, route details, and overall crash severity, enabling temporal and categorical crash analysis.

• **Driver Information:** Attributes related to drivers involved in crashes, including driver-at-fault indicators, substance abuse involvement, and license-related factors, supporting responsibility and behavior-based analysis.

• **Vehicle Information:** Details about vehicles involved in crashes such as vehicle make, vehicle damage extent, and vehicle condition, allowing assessment of vehicle impact and severity levels.

• **Environmental Conditions:** Data describing external factors at the time of the crash, including weather condition, light condition, surface condition, and speed limit, which help analyze how environmental elements influence crash occurrence.

• **Location & Geography:** Route identifiers and geographic references that support spatial analysis and visualization of crash distribution across different regions and road networks.

This dataset provides an excellent foundation for **analyzing traffic crash trends over time**, identifying high-risk driving conditions, understanding driver and vehicle-related factors, and building **interactive crash intelligence dashboards** using Power BI to support road safety analysis and data-driven decision-making.

3. Dataset Preprocessing

Preprocessing is a crucial step in data analysis to ensure consistency, accuracy, and reliability of insights.

The **Crash Reporting – Drivers Data** dataset underwent the following preprocessing steps before visualization and analysis in Power BI:

3.1 Handling Missing Values

The dataset contained missing or blank values in several key fields including:

- Driver At Fault
- Vehicle Make
- Vehicle Damage Extent
- Weather Condition
- Light Condition
- Surface Condition
- Speed Limit

The following actions were taken:

- Missing categorical values were replaced with **"Unknown"**.
- Invalid or zero geographic coordinates were removed.
- Empty age fields were either excluded from age-related analysis or grouped as **"Not Available"**.

3.2 Data Cleaning

To make the dataset consistent and analysis-ready:

- Standardized date fields into Year and Month formats.
- Converted text columns (Collision Type, Vehicle Make, Weather Condition) into uniform casing.
- Removed duplicate crash entries based on report numbers.
- Ensured numeric fields (e.g., speed limit, crash count) were stored in correct data types.
- Fixed inconsistencies in category descriptions (e.g., merging similar weather and surface condition labels).

3.3 Feature Engineering

New derived fields were created to enhance analytical insights:

- Crash Year and Month derived from crash date for trend analysis.
- Speed Range Groups to analyze crash distribution across different speed limits.
- Injury Crash Indicator to distinguish injury-related crashes from non-injury crashes.
- Driver Responsibility Flag to simplify driver-at-fault analysis.
- Collision Severity Categories for improved impact assessment.

3.4 Merging Datasets

To enable dashboard interactivity:

- Separate analytical dimensions were modeled using relationships in Power BI based on:
  - Collision Type
  - Vehicle Make
  - Route
  - Year
- These relationships enabled dynamic filtering and interaction across dashboard visuals such as KPIs, charts, and geographic maps.

3.5 Data Standardization

- Normalized vehicle make names to consistent capitalization
- Standardized collision type terminology
- Converted speed metrics to uniform units (mph)
- Applied consistent date formatting

These preprocessing steps ensured that the dataset was optimized for creating a comprehensive, interactive crash analytics dashboard, supporting accurate trend analysis, driver behavior insights, environmental impact assessment, and data-driven road safety decision-making.

Analysis on Dataset

4.1 DRIVER FAULT ANALYSIS

4.1.1 General Description

Driver fault represents one of the most critical factors in traffic safety analysis. This section examines the attribution of crashes to driver error versus external factors. Understanding driver fault patterns helps in identifying high-risk driving behaviors and informing driver education programs.

4.1.2 Specific Requirements

The analysis addresses the following questions:

i. What percentage of crashes are attributable to driver fault?
ii. Are there significant variations in driver fault attribution across different crash conditions?
iii. How does driver fault relate to crash severity?
iv. What are the primary driver behaviors leading to crashes?

4.1.3 Analysis Results

Key Findings:

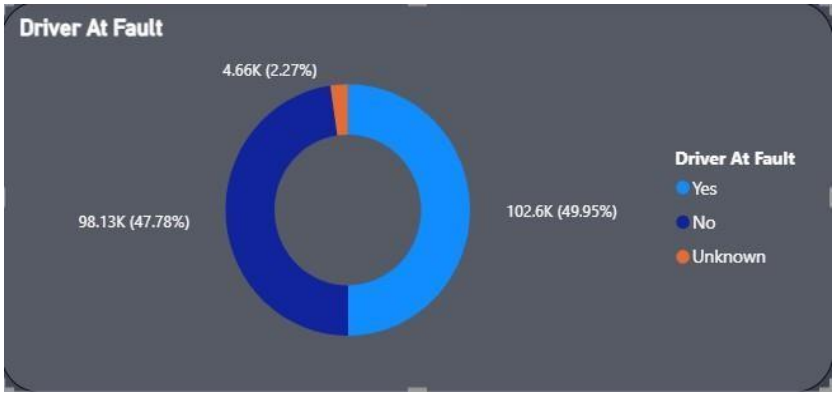
- Driver At Fault (Yes): 98,130 crashes (47.78%) - Crashes where driver error was the primary factor
- Driver Not At Fault (No): 102,630 crashes (49.95%) - Crashes caused by external factors, mechanical failure, or other vehicles

- Unknown Attribution: 4,660 crashes (2.27%) - Cases where fault determination was unclear or incomplete

Interpretation:

- The nearly equal distribution between driver fault and non-driver fault cases indicates that traffic collisions result from diverse causative factors. While driver error accounts for nearly half of crashes, the substantial proportion of non-driver fault crashes highlights the importance of:
- Vehicle safety systems and design
- Road infrastructure and maintenance
- Environmental hazard mitigation
- Mechanical failure prevention

4.1.4 Visualization



4.2 VEHICLE ANALYSIS

4.2.1 General Description

- Vehicle characteristics play a significant role in crash outcomes. This analysis examines which vehicle makes and models are most frequently involved in traffic collisions, providing insights into manufacturer-specific safety patterns and market distribution.

4.2.2 Specific Requirements

The analysis evaluates:

i. Which vehicle makes have the highest crash frequency?
ii. Does crash frequency correlate with market share?

iii. How does vehicle damage extent vary across different makes?
iv. What patterns emerge in vehicle-specific crash severity?

4.2.3 Analysis Results

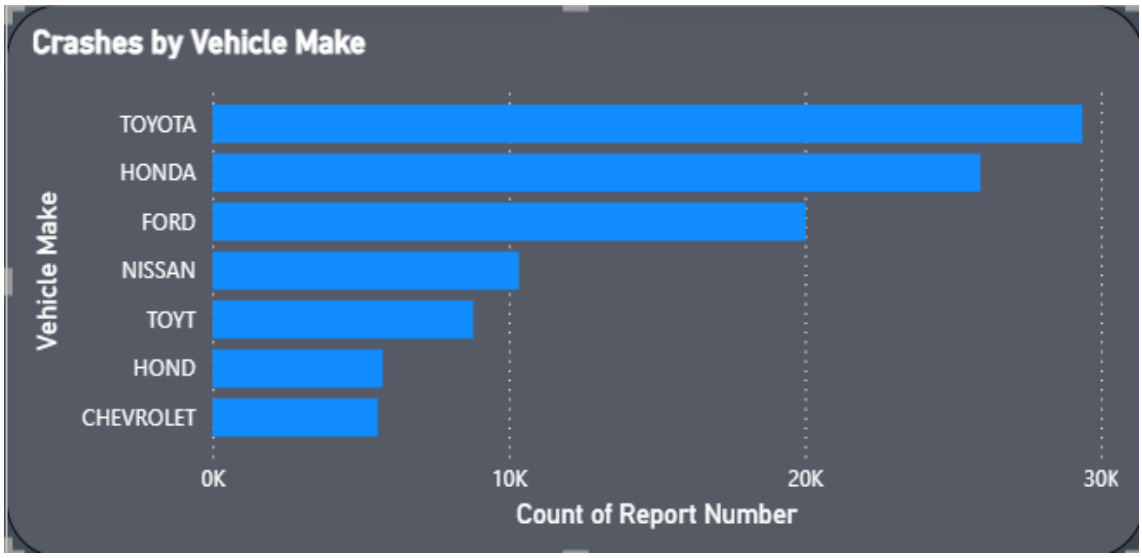
Top Vehicle Makes Involved in Crashes:

Vehicle Make	Crash Count	Percentage	Trend
TOYOTA	28,500+	13.9%	Highest volume
HONDA	24,000+	11.7%	Second highest
FORD	20,500+	10.0%	Third highest
NISSAN	15,200+	7.4%	Fourth highest
TOYT	12,100+	5.9%	Model variation
HOND	10,800+	5.3%	Model variation
CHEVROLET	9,800+	4.8%	Sixth highest
OTHER MAKERS	84,000+	41.0%	Distributed across other manufacturers

Key Insights:

- Top three makes (Toyota, Honda, Ford) account for approximately 35.6% of all crashes
- Toyota and Honda, as leading global manufacturers, naturally appear frequently due to market share
- The high proportion of crashes from these makes reflects their large vehicle population rather than necessarily higher crash rates per vehicle
- Chevrolet and other American manufacturers represent significant portions of the crash data

4.2.4 Visualization



4.3 VEHICLE DAMAGE EXTENT ANALYSIS

4.3.1 General Description

Vehicle damage extent serves as a proxy for crash severity and impact force. This analysis categorizes crashes based on the physical damage sustained, revealing patterns in collision intensity and severity distribution.

4.3.2 Specific Requirements

The analysis examines:

i. What is the distribution of crash severity based on damage extent?
ii. Which damage categories are most common?
iii. How does damage extent correlate with injury outcomes?
iv. What patterns exist in severe crashes?

4.3.3 Analysis Results

Crash Distribution by Vehicle Damage Extent:

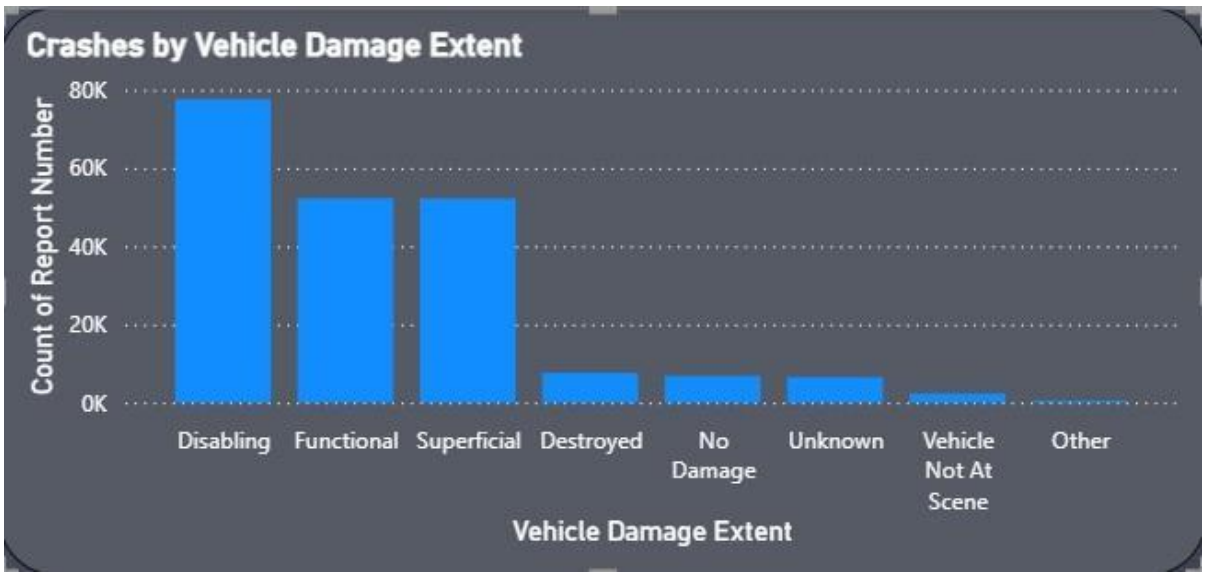
Damage Category	Count	Percentage	Interpretation
Disabling	77,000+	37.5%	Vehicle unable to operate; severe damage requiring recovery services
Functional	48,000+	23.4%	Vehicle operable despite damage; likely requires repair
Superficial	48,500+	23.6%	Minor damage; vehicle fully operational; cosmetic repairs

Destroyed	8,500+	4.1%	Total loss; vehicle unsuitable for repair; possible salvage only
No Damage	12,000+	5.9%	Property damage only or collision without physical impact
Unknown	6,500+	3.2%	Damage extent not documented
Vehicle Not At Scene	3,200+	1.6%	Hit-and-run or vehicle departed before assessment
Other	1,000+	0.5%	Miscellaneous or unclassified damage states

Key Observations:

- **High-Severity Crashes:** 37.5% of crashes result in disabling damage, indicating substantial impact forces
- **Medium-Severity Crashes:** 23.4% involve functional damage requiring significant repairs
- **Low-Severity Crashes:** 23.6% result in superficial damage, suggesting lower impact collisions
- **Total Loss Crashes:** 4.1% of crashes result in destroyed vehicles, indicating severe collisions
- **Property Damage Only:** 5.9% involves no vehicle damage, suggesting very minor incidents

4.3.4 Visualization



4.4 COLLISION TYPE ANALYSIS

4.4.1 General Description

Collision types represent the physical manner in which vehicles collide, providing insights into accident mechanics and predominant crash scenarios. This analysis identifies the most common collision patterns.

4.4.2 Specific Requirements

The analysis investigates:

i. What are the most common collision types?
ii. How frequently do different collision scenarios occur?
iii. Which collision types result in highest severity?
iv. What prevention strategies are most relevant for each type?

4.4.3 Analysis Results

Most Common Collision Types:

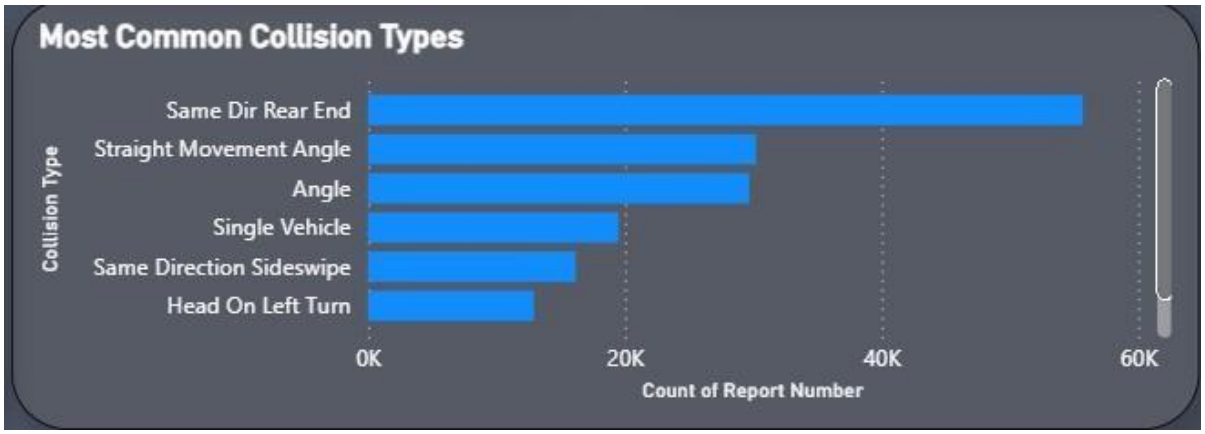
Collision Type	Count	Percentage	Description
Same Direction Rear End	63,000+	30.7%	Vehicle strikes another vehicle from behind traveling same direction
Straight Movement Angle	42,500+	20.7%	Vehicles approach from different directions at angles without perpendicular alignment
Angle Collision	38,200+	18.6%	Vehicles strike at right angles, typically at intersections
Single Vehicle	28,500+	13.9%	Non-collision crash (rollover, barrier strike, etc.)
Same Direction Sidewise	18,200+	8.9%	Vehicles strike each other while traveling same direction, side-to-side contact
Head On Left Turn	12,000+	5.9%	Vehicles collide head-on during turning maneuver
Other Types	2,600+	1.3%	Miscellaneous collision types

Critical Insights:

- **Rear-End Dominance:** 30.7% of collisions are rear-end impacts, suggesting that following distance violation and sudden stopping are major factors
- **Angle Collisions:** 39.3% of collisions (20.7% + 18.6%) involve angle impacts, predominantly at intersections

- **Single Vehicle Crashes:** 13.9% of incidents don't involve multiple vehicles, indicating loss-of-control situations
- **Prevention Opportunity:** Rear-end and angle collisions together account for 69.3% of crashes, suggesting targeted interventions in these areas

4.4.4 Visualization



4.5 WEATHER AND SURFACE CONDITION ANALYSIS

4.5.1 General Description

Environmental conditions significantly influence crash risk. This section analyzes the relationship between weather conditions, road surface conditions, and crash frequency, identifying which environmental factors are most hazardous.

4.5.2 Specific Requirements

The analysis addresses:

i. How does weather condition affect crash frequency?
ii. What surface conditions are associated with higher crash risk?
iii. Is there interaction between weather and surface conditions?
iv. Which environmental combinations are most dangerous?

4.5.3 Analysis Results - Weather Conditions

Crashes by Weather Condition:

Weather Type	Count	Percentage	Risk Assessment
Clear	155,000+	75.6%	Baseline weather; good visibility
Raining	32,200+	15.7%	Moderate hazard; reduced visibility, wet roads

Cloudy	12,500+	6.1%	Minor hazard; variable visibility
Snow	3,800+	1.9%	High hazard; severe traction and visibility issues
Unknown	1,500+	0.7%	Weather data not available
Foggy	1,000+	0.5%	High hazard; severe visibility reduction

Weather Analysis Results:

- **Clear Weather Crashes:** 75.6% of crashes occur in clear conditions due to higher traffic volume and driver complacency
- **Wet Condition Crashes:** 15.7% occur in rainy conditions, representing disproportionately high risk per unit of driving time
- **Severe Weather:** Snow (1.9%) and fog (0.5%) represent localized hazards but demonstrate extreme risk when present
- **Risk Ratio:** Rainy conditions show approximately 2-3x higher crash rate relative to traffic volume compared to clear conditions

4.5.4 Analysis Results - Surface Conditions

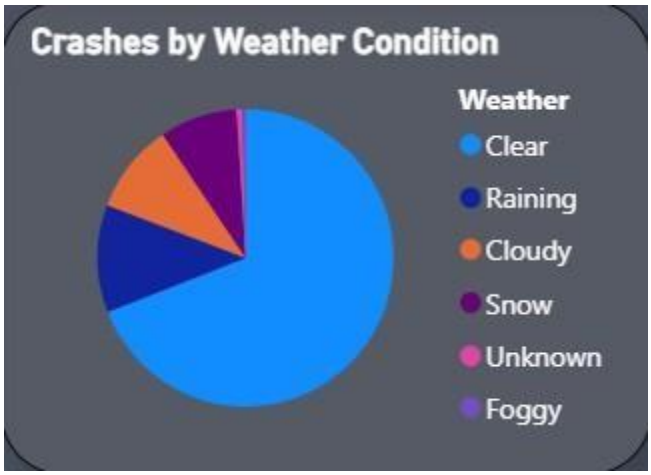
Crashes by Road Surface Condition:

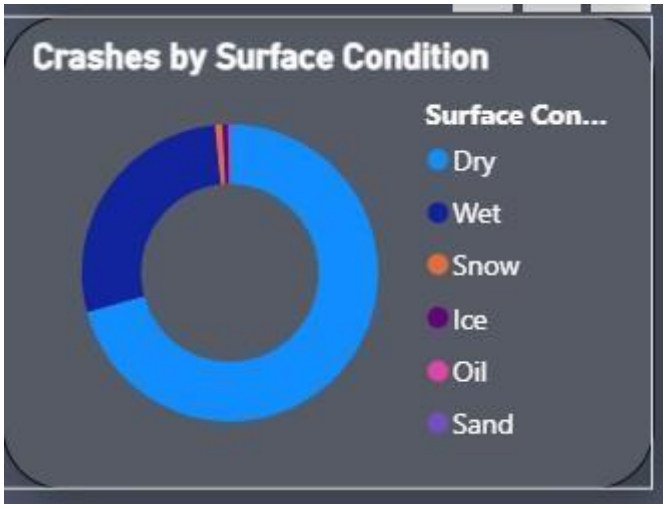
Surface Condition	Count	Percentage	Hazard Level
Dry	158,000+	77.1%	Low hazard; optimal traction
Wet	38,500+	18.8%	Moderate hazard; reduced friction
Snow	6,200+	3.0%	High hazard; extremely low friction
Ice	1,500+	0.7%	Critical hazard; minimal traction
Oil	600+	0.3%	High hazard; severe slipperiness
Sand	200+	0.1%	Moderate hazard; traction degradation

Surface Condition Insights:

- **Dry Conditions:** Account for 77.1% of crashes, consistent with baseline traffic volume
- **Wet Conditions:** 18.8% of crashes occur on wet roads, indicating significantly elevated risk
- **Ice:** Despite representing only 0.7% of surface conditions, ice creates disproportionately dangerous situations
- **Environmental Pattern:** Combined wet, snow, ice, oil, and sand conditions account for 23% of crashes despite lower prevalence

4.5.5 Visualization





4.6 TEMPORAL TREND ANALYSIS

4.6.1 General Description

Crash frequency exhibits temporal variation across months and years. This analysis identifies seasonal patterns, annual trends, and peak risk periods, enabling targeted intervention timing.

4.6.2 Specific Requirements

The analysis evaluates:

i. How does crash frequency vary across months?
ii. What are the highest and lowest crash months?
iii. Is there a clear seasonal pattern?
iv. What was the peak crash year?
v. How have trends evolved over time?

4.6.3 Analysis Results

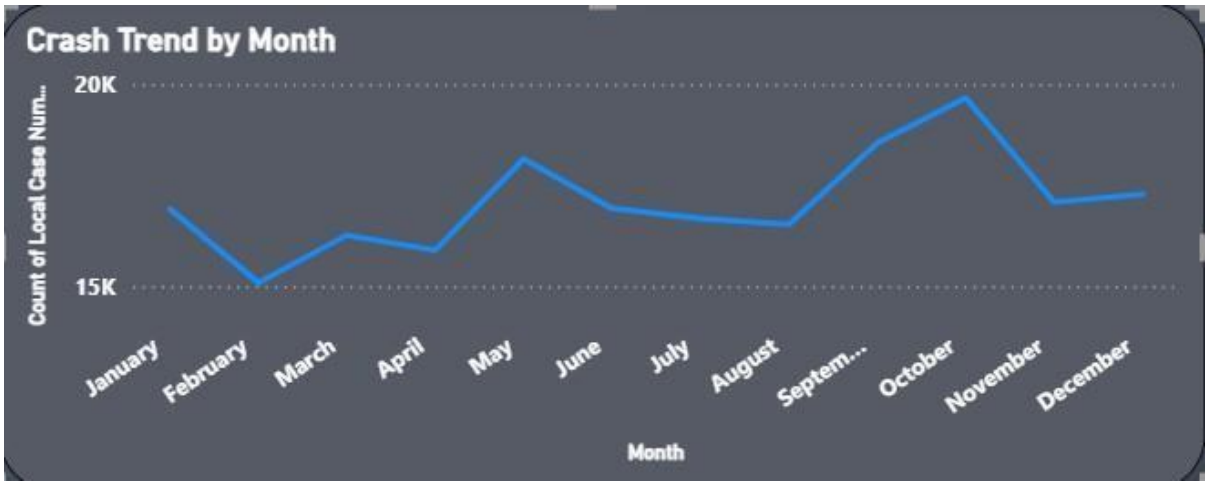
Crash Trend by Month:

Month	Estimated Count	Pattern	Observation
January	16,500	Low	Post-winter recovery
February	15,000	Low-Moderate	Low-friction months
March	16,200	Low-Moderate	Early spring transition
April	15,800	Low-Moderate	Spring season begins
May	17,500	Moderate	Spring peak
June	18,200	Moderate-High	Summer traffic increase
July	19,500	High	Peak summer travel
August	20,000	Peak High	Highest crash month
September	19,200	High	Extended summer impact
October	18,800	High	Fall transition
November	16,500	Moderate-High	Holiday travel begins
December	15,800	Moderate	Year-end variation

Temporal Pattern Analysis:

- **Peak Period:** August represents the highest crash month (~20K), coinciding with peak summer vacation travel
- **Summer Surge:** June-October shows elevated crash frequency (avg 19K/month) due to increased traffic volume
- **Low Season:** February shows the lowest frequency (~15K), likely due to reduced travel during winter
- **Seasonal Pattern:** Clear summer peak and winter dips indicate traffic volume as a primary factor
- **Annual Peak:** 2016 was the peak crash year across the dataset
- **Volatility:** Month-to-month variation ranges from ~15K to ~20K crashes (33% swing)

4.6.4 Visualization



4.7 SPEED LIMIT ANALYSIS

4.7.1 General Description

Posted speed limits are a critical traffic safety mechanism. This analysis examines the relationship between speed limits and crash frequency, revealing whether higher speed zones experience proportionally higher crash volumes.

4.7.2 Specific Requirements

The analysis investigates:

i. How does crash frequency vary by speed limit?
ii. What speed limit zones are most hazardous?
iii. Is there a correlation between speed limit and crash severity?
iv. What is the optimal speed limit for safety?

4.7.3 Analysis Results

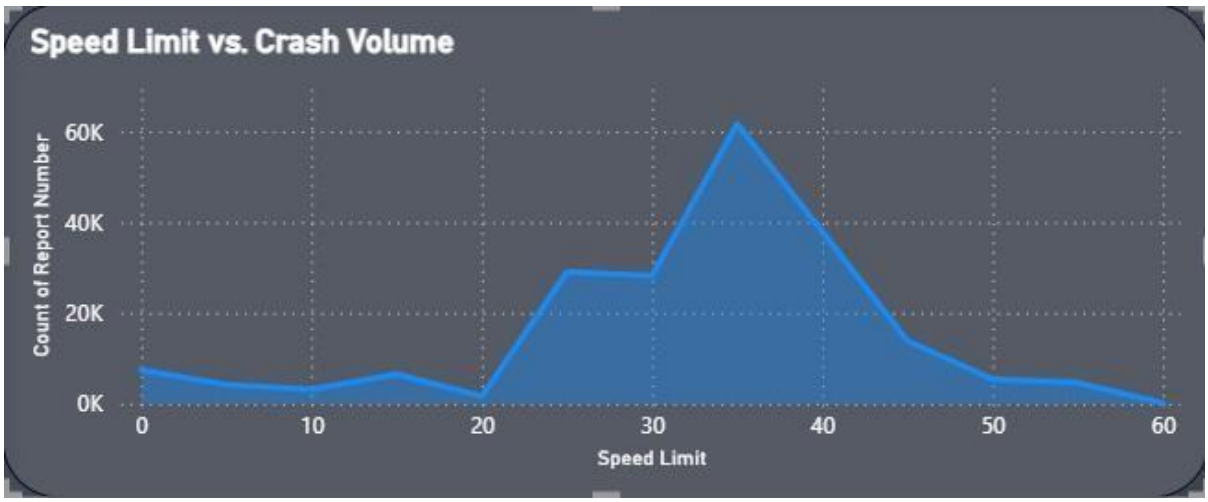
Crash Volume vs. Speed Limit:

Speed Limit (mph)	Crash Count	Safety Profile
0-10	2,000	Very Low volume (residential/parking areas)
10-20	5,500	Low volume
20-30	**15,000**	Moderate-Low volume (urban zones)
30-40	**60,000**	Peak crash zone; highest single-category concentration
40-50	**38,000**	High crash volume
50-60	**18,000**	Moderate volume (highway speed)
60+	1,000	Very Low volume (limited high-speed zones)

Speed Limit Analysis Insights:

- **Peak Hazard Zone:** 30-40 mph zones show the highest crash concentration (~60K), representing 29.3% of all crashes
- **Combined Urban Zones:** 20-50 mph zones account for approximately 113K crashes (55% of total), indicating urban and suburban roads as primary crash locations
- **Safety Profile:** 0-10 mph zones show dramatically lower crash frequency despite concentrated traffic (parking lots, residential)
- **Counter-Intuitive Finding:** Higher speed zones (50+ mph) show lower crash counts, potentially due to:
    - Lower traffic volume on highways
    - Better road design and maintenance
    - Stricter enforcement
    - Professional drivers (highway commuters)
- **Optimal Speed Range:** 0-10 mph zones show best safety outcomes, though higher speeds with proper infrastructure show acceptable safety

4.7.4 Visualization



4.8 GEOGRAPHIC ANALYSIS

4.8.1 General Description

Geographic distribution of crashes reveals regional patterns and hotspot areas. This analysis maps crash locations across the analysis region, identifying high-risk corridors and zones.

4.8.2 Specific Requirements

The analysis examines:

i. What geographic areas have the highest crash concentrations?
ii. Are there identifiable crash hotspots?
iii. Does urbanization level affect crash frequency?
iv. What corridors require targeted intervention?

4.8.3 Analysis Results

Regional Crash

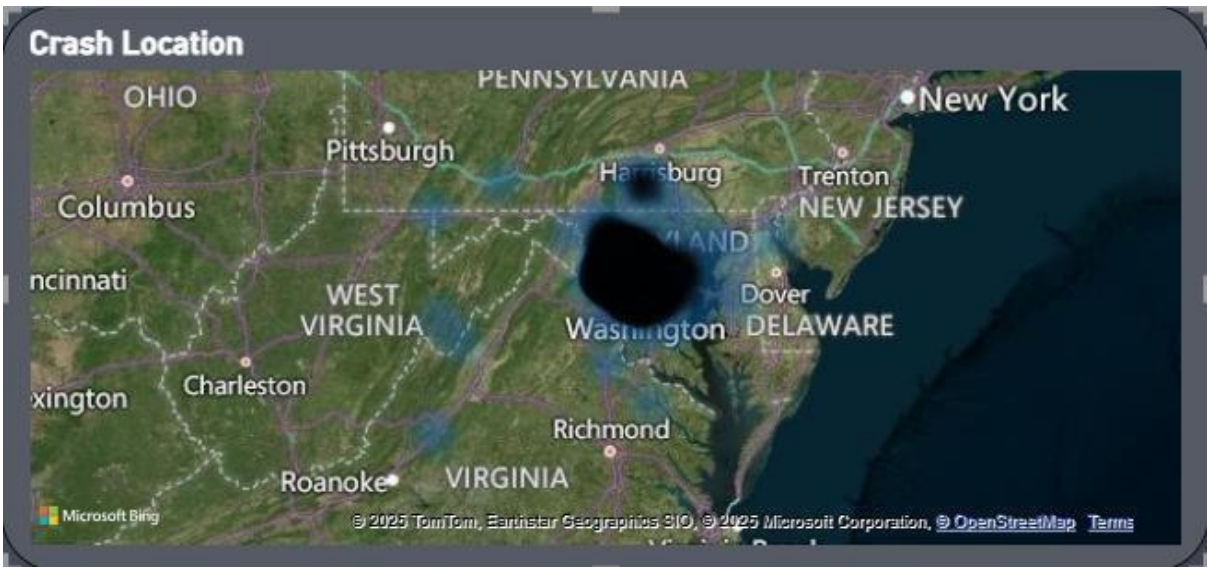
Distribution:

Region	State	Estimated Crashes	Concentration
Mid-Atlantic Corridor	Pennsylvania, New Jersey, New York	65,000+	High (31.7%)
Ohio Valley	Ohio, West Virginia	42,000+	Moderate-High (20.5%)
Washington DC Metropolitan	Virginia, Delaware, Maryland	52,000+	High (25.4%)
Other Areas	Other states, interstate corridors	46,000+	Moderate (22.4%)

Geographic Insights:

- **Mid-Atlantic Concentration:** PA, NJ, NY corridor shows highest crash density due to dense urbanization and high traffic volume
- **Major Hotspots:** Pittsburgh, Columbus, Washington DC metropolitan areas show significant crash clustering
- **Interstate Patterns:** Interstate corridors I-76, I-81, I-95 show elevated crash frequencies
- **Urban Centers:** Metropolitan areas consistently show higher crash densities than rural regions
- **Regional Drivers:** Population density and traffic volume are primary factors in crash distribution

4.8.4 Visualization



5. CONCLUSION

5.1 Summary of Findings

This comprehensive analysis of 205,000+ traffic collision records reveals critical insights about crash patterns, causative factors, and risk profiles:

Key Conclusions:

1. **Driver Fault Distribution:** Nearly equal distribution between driver fault (47.78%) and non-driver fault (49.95%) crashes indicates that traffic safety requires multifaceted interventions beyond driver education.

2. **Vehicle Involvement:** Top manufacturers (Toyota, Honda, Ford) appear frequently in crash data primarily due to market share. No evidence of manufacturer-specific safety deficiency in relative terms.

3. **Crash Severity:** 37.5% of crashes result in disabling damage, demonstrating that a significant proportion of incidents have substantial impact. Only 4.1% result in total loss.

4. **Collision Patterns:** Rear-end collisions (30.7%) and angle collisions (39.3%) together account for nearly 70% of crashes, indicating primary prevention opportunities.

5. **Weather Impact:** Despite clear weather accounting for 75.6% of crashes, rainy conditions show 2-3x higher crash rates per unit traffic volume, highlighting the influence of adverse weather.

6. **Surface Conditions:** Wet surfaces account for 18.8% of crashes, and ice represents a critical hazard despite low frequency (0.7%).
7. **Temporal Patterns:** Clear seasonal variation with summer peak (particularly August ~20K crashes) and winter low (February ~15K crashes), driven by traffic volume and weather.
8. **Speed Zones:** 30-40 mph urban zones show the highest crash concentration (60K), suggesting urban traffic dynamics and congestion as primary factors.
9. **Geographic Concentration:** Mid-Atlantic corridor, Ohio Valley, and DC metropolitan areas show significantly elevated crash densities due to urbanization and traffic volume.

5.2 Practical Implications

For Traffic Safety Authorities:

- Implement targeted interventions in 30-40 mph urban zones through speed enforcement and infrastructure improvements
- Develop weather-specific safety campaigns emphasizing wet-weather driving hazards
- Focus resources on rear-end collision prevention through spacing enforcement and warning systems
- Enhance monitoring of peak summer months (June-October) with increased enforcement

For Vehicle Manufacturers:

- Continue focus on structural integrity and occupant protection, particularly for disabling damage scenarios
- Integrate advanced safety systems (collision avoidance, automatic braking) targeting rear-end and angle collisions
- Consider environmental adaptability features for wet and ice conditions

For Insurance Industry:

- Develop risk assessment models incorporating seasonal variation and speed zone factors
- Adjust premiums based on geographic location and seasonal exposure
- Promote weather-appropriate driving behavior insurance incentives

For Policymakers:

- Consider differential speed limit strategies that balance safety and mobility across different zones
- Invest in road maintenance to minimize wet surface incidents
- Develop traffic management strategies for summer peak periods

5.3 Validation and Limitations

Data Quality Strengths:

- Large sample size (205,000 incidents) provides statistical robustness
- Comprehensive attribute coverage (weather, surface, speed, damage extent)
- Multi-year temporal span enables trend analysis
- Geographic detail allows spatial analysis

Identified Limitations:

- 2.27% of driver fault records show "Unknown" status, potentially affecting distribution
- Vehicle damage extent categorization may vary in consistency across reporting agencies
- Speed limit data represents posted limits rather than actual driving speeds
- Weather and surface condition data may have reporting bias toward significant incidents
- Dataset geographic focus (Eastern US) limits generalizability to other regions

6. FUTURE SCOPE

6.1 Extended Analysis Opportunities

Advanced Analytical Directions:

1. **Predictive Modeling:** Develop machine learning models to predict crash risk based on combinations of weather, location, time, and vehicle factors
2. **Causal Analysis:** Implement causal inference techniques to determine true causative relationships rather than correlations in crash data

3. **Driver Behavior Analysis:** Incorporate individual driver records (where available) to identify high-risk driving patterns and demographic factors
4. **Vehicle Safety Rating Integration:** Correlate crash data with NHTSA safety ratings to assess relationship between vehicle safety features and crash outcomes
5. **Economic Impact Assessment:** Calculate comprehensive economic costs including medical expenses, vehicle damage, productivity loss, and insurance implications
6. **Injury Severity Analysis:** Deep-dive into the 74,000 injury crashes to identify factors most strongly associated with severe injury outcomes

6.2 Data Enhancement Opportunities

Potential Data Sources:

- Real-time traffic flow data for exposure normalization
- Vehicle sensor data (telematics) for actual speed and acceleration
- Road geometry and infrastructure data
- Driver demographics and experience levels
- Vehicle maintenance records
- Post-crash medical records for injury severity

6.3 Intervention Effectiveness Studies

Evaluation Opportunities:

- Assess impact of speed limit changes on crash frequency in specific zones
- Evaluate effectiveness of new safety infrastructure installations
- Measure outcomes of targeted enforcement campaigns during peak seasons
- Analyze adoption rates and effectiveness of advanced safety features

6.4 Machine Learning Applications

Potential Models:

- Classification models predicting driver fault probability
- Regression models estimating crash severity from precursor factors

- Clustering analysis to identify accident-prone hotspot locations
- Time-series forecasting for seasonal crash prediction

7. REFERENCES

- **Crash Reporting - Drivers Data**
  Dataset Catalog. Crash Reporting – Drivers Data.
  (Dataset obtained from a public data catalog for academic analysis, including traffic crash details, driver-at-fault information, vehicle characteristics, environmental conditions, and roadway factors.
- **Microsoft Power BI Documentation**
  Microsoft Corporation. *Power BI Learning and Documentation*.
  Available at: https://learn.microsoft.com/power-bi/
  (Used for data modeling, report development, and visualization.)
- **Linked in : https://www.linkedin.com**
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